

Lennart Levin Vetter

Machine Learning Engineer

+49 171 9800 458 | Leipzig, Germany | len.vetter@gmail.com | llmvetter.github.io

SUMMARY

Machine Learning Engineer with 2.5 years of experience in building scalable Classification and Recommendation microservices that deliver measurable business impact. Backed by strong research experience with Diploma Thesis in Inverse Reinforcement Learning supervised by Prof. Dr. Ostap Okhrin. Proven ability to translate complex research papers into production ready code. Strategic business mindset due to diverse background in Venture Capital and Startup Consulting.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer

Apr 2024 - Aug 2025

35up GmbH

Remote

- Architected Product Classification Service (5.500+ classes) combining Vector Search with Cross-Encoder based reranking, achieving 91% accuracy and fully automating manual tagging efforts.
- Designed Product Recommendation Service based on graph traversal (category discovery) and vector search (product retrieval) scaling AI based recommendations from 0% to 25%.
- Architected Product Attribute Prediction Service to predict arbitrary attributes by leveraging LLM structured generation and Pydantic.
- Outlined strategic ML roadmap for fully AI based recommendation pipeline with CTO. Executed prototyping, development and integration of ML microservices (Classification, Attribute Prediction, Recommendation).

Data Scientist

Jan 2023 - April 2024

35Up GmbH

Remote

- Benchmarked SOTA embedding models and proprietary models to find optimal cost/accuracy/latency tradeoff for production deployment.
- Engineered data pipelines and synthetic dataset generation scripts to train and evaluate supervised models.
- Automated web scraping and data annotation workflows, reducing manual data processing efforts for core ML team.

Graduate Researcher - Inverse Reinforcement Learning

Aug 2023 - June 2024

Reinforcement Learning Research Group Dresden

Dresden, Germany

- Conducted research on Inverse Reinforcement Learning applied to Autonomous Driving, focusing on Maximum Entropy IRL and Inverse Q-Learning to model car following behavior.
- Engineered custom stochastic RL environment (OpenAI Gym) including state space definition and transition model, incorporating car physics and probabilistic transition dynamics
- Translated research papers into PyTorch code, trained RL models on slurm based GPU cluster.
- Validated reward function extraction from expert demonstrations, proving that resulting RL agents are able to mimic human driving behaviour successfully

Visiting Investment Analyst

Jan 2022 - April 2022

Capnamic Ventures GmbH

Berlin, Germany

- Conducted technical and financial due diligence on ~60 B2B SaaS investment opportunities per week, identifying valuable business cases for Partners.
- Performed hypothesis-driven market research and validated business cases through financial modelling (cap tables, dilution simulations)

SELECTED SKILLS

Machine Learning: Deep Learning, Transformers, LLMs, Embeddings, NLP, Reinforcement Learning

Frameworks: JAX, PyTorch, Pydantic, FastAPI, OpenAI Gym

Data: MongoDB, Milvus (Vector), Neo4J/Neptune (Graph), Information Retrieval (RAG, MMR, Reranking)

Infrastructure: GCP, AWS, Terraform, Slurm

DevOps: CI/CD, Microservice Architecture, Docker, Kubernetes

EDUCATION

Diploma, Industrial Engineering | Advanced Data Analytics

2026

Technical University Dresden

Dresden, Germany

MS, Machine Learning

2025

The Cooper Union (Exchange)

New York City, USA

MS, Data Science

2023

Université Paris Nanterre (Exchange)

Paris, France